



Module Title

Cybersecurity and Networking

Internship Assignment Submission

OWASP Top 10:2025 Risk Identification

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1.ASSIGNMENT OWASP TOP 10:2025 — RISK IDENTIFICATION

1.1 Answer all questions in your own words.

1.1.1 PART A — FUNDAMENTALS

1.What does OWASP stand for?

OWASP stands for Open Worldwide Application Security Project. It is a global nonprofit organization focused on improving software and application security.

2. What is the purpose of the OWASP Top 10?

The OWASP Top 10 explains the major categories of web application security risks. It helps developers and security professionals understand and prevent common security problems.

3. Why is the OWASP Top 10 useful for cybersecurity students?

It helps students learn common security risks and understand important cybersecurity terms. It also provides a foundation for security testing and secure application development.

4. What is a vulnerability?

A vulnerability is a weakness or security flaw in system that could be misused or cause security problems.

5. What is security risk?

Security risk is the possibility of harm or loss caused by a security weakness. It considers both the likelihood of a problem and its possible impact.

6. What is an asset?

An asset is something valuable that needs protection. Examples include user accounts, personal information, databases and source code.

7. What is a threat?

A threat is something that can potentially cause harm by taking advantage of a security weakness.

8. What is impact?

Impact is the possible consequences of a security problem. It can include data loss. Unauthorized access, privacy problems, or service disruption.

9. Explain Authentication.

Authentication is the process of verifying who a user is. For example, entering a username and password to login.

10. Explain Authorization.

Authorization checks what an authenticated user is allowed to access or do. For example, checking whether a user can access an admin panel.

11. What is the difference between Authentication and Authorization?

Authentication checks who you are, while Authorization checks what you are allowed to do.

Example: Login verifies your identity (authentication), and access to specific files depends on your permissions (authorization).

1.1.2 PART B — OWASP TOP 10:2025

Write one or two sentences explaining each category:

12. A01:2025 — Broken Access Control

Broken Access Control occurs when users can access data or functions, they are not allowed to access. Example: a normal user accessing an admin page.

13. A02:2025 — Security Misconfiguration

Security Misconfiguration occurs when security settings are incorrect or unsafe. Example: displaying detailed error information to users.

14. A03:2025 — Software Supply Chain Failures

Security Misconfiguration occurs when security settings are incorrect or unsafe. Example: displaying detailed error information to users.

15. A04:2025 — Cryptographic Failures

Cryptographic Failures occur when sensitive data is not properly protected using encryption or other security controls. This can expose confidential information.

16. A05:2025 — Injection

Injection occurs when untrusted input is treated as part of a command or query. SQL Injection is a common example.

17. A06:2025 — Insecure Design

Insecure Design occurs when security is not properly considered during application design. Example: a sensitive action without proper authorization or approval controls.

18. A07:2025 — Authentication Failures

Authentication Failures occur when an application does not properly verify or protect user identities. Example: weak login or password controls.

19. A08:2025 — Software or Data Integrity Failures

This occurs when software or data is not properly verified for integrity or trustworthiness. Example: accepting an update without checking its source.

20. A09:2025 — Security Logging and Alerting Failures

This occurs when important security events are not properly logged or alerted. As a result, suspicious activity may go unnoticed.

21. A10:2025 — Mishandling of Exceptional Conditions

This occurs when unexpected errors or situation are not handled safely. Example: an error causes an important security check to be skipped.

1.1.3 PART C — SAFE PRACTICAL ANALYSIS Analyze these FICTIONAL scenarios.

SCENARIO 1: A normal student can access an admin-only page.

Identify: Asset: Admin functions and student information.

Security Problem: A normal student can access a page that should only be available to administrators.

OWASP Category: A01:2025 – Broken Access Control

Possible Impact: The student gain unauthorized access to restricted information or functions.

High-Level Defensive Recommendation: Implement and properly enforce server-side authorization checks for every restricted function and page.

SCENARIO 2: A production application displays internal file paths and detailed debug errors.

OWASP Category: A02:2025 – Security Misconfiguration

Why: The application is incorrectly configured to reveal unnecessary technical information to users. Production systems should use safe error handling and avoid exposing internal details.

SCENARIO 3: A company unknowingly installs a compromised third-party software dependency.

OWASP Category: A03:2025 – Software Supply Chain Failures

Why: The security problem comes from a third-party software dependency that has been compromised. Software dependencies should be properly reviewed and managed.

SCENARIO 4: A sensitive approval workflow was designed without an approval control.
Identify:

OWASP Category: A06:2025 – Insecure Design

Why: The required security control was missing from the applications design. Security should be considered when designing sensitive workflows.

SCENARIO 5: An application does not record important suspicious authentication activity.
Identify:

OWASP Category: A09:2025 – Security Logging and Alerting Failures

Why: Important security activity is not being properly logged or monitored. This can make it difficult to detect and investigate suspicious authentication events.

1.1.4 PART D — STUDENT PORTAL ASSETS Imagine a fictional internship portal.
List five important assets.

Asset 1: Student Accounts

Why it needs protection: Student account contains login information and personal access. Protecting them helps prevent unauthorized account access.

Asset 2: Assignment Submission

Why it needs protection: Assignments are students' academic work and should not be modified, deleted or accessed by unauthorized users.

Asset 3: Personal Information

Why it needs protection: Personal Information is private and could cause privacy problems if accessed or exposed without permission.

Asset 4: Internship Certificates

Why it needs protection: Certificates are important record that may be used a proof of training or internship completion. They should be protected from unauthorized modification or deletion.

Asset 5: Application Database:

Why it needs protection: The database may contain student record, account information, and other important data. Protecting it helps prevent unauthorized access, modification, or data loss.

